

MEAN - VARIANCE BASED PORTFOLIO OPTIMISATION

In the Indian Stock Market



Enrolment No - 24114022

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3rd Year, CSE



Project Overview

Sector-Specific Portfolio Optimization

The Indian Stock Market is heavily influenced by major industry sectors such as **IT, Agriculture, FMCG, Pharmaceuticals, Energy, etc.** This makes the Indian stock market diversified and therefore complex. Portfolio Optimization has been a challenging problem that has attracted the attention of many analysts. It is particularly difficult due to the volatile nature of the economy and industries.

INDEX NAME	INDEX VALUE ON DEC. 31, 2019	INDEX VALUE ON DEC. 31, 2020	ABSOLUTE RETURNS (%)
S&P BSE Healthcare	14,998.90	24,390.31	62.61
S&P BSE Information Technology	19,724.29	31,568.43	60.05
S&P BSE Basic Materials	3,425.11	4,406.44	28.65
S&P BSE Consumer Discretionary	4,309.69	5,233.94	21.45
S&P BSE Industrials	3,369.48	4,025.04	19.46
S&P BSE Energy	6,736.88	8,044.64	19.41
S&P BSE Telecom	1,234.40	1,413.24	14.49
S&P BSE FMCG	14,573.04	16,495.37	13.19
S&P BSE Realty	2,487.72	2,716.65	9.20
S&P BSE Utilities	2,325.76	2,423.00	4.18
S&P BSE Finance	8,403.44	8,508.25	1.25

Source: S&P Dow Jones Indices LLC. Data from Dec. 31, 2019, to Dec. 31, 2020. Past performance is no guarantee of future results. Table is provided for illustrative purposes.

Performance of various BSE Sectors during the period 2019-2020

Thus, a mix of stocks from every sector reduces the overall risk in the portfolio and enhances the chances of having a larger return. With the help of the model of **Mean Variance Optimization**, we will find the portfolio with stocks from various sectors that has the minimized risk for a given target return.



Project Deliverables

I have prepared and delivered the following things throughout my project:

- (a) Sectoral statistics including **volatility, average returns, correlation matrices and beta values** of every stock within a sector
- (b) Calculation of **Sharpe and Sortino ratios** of all stocks in a sector
- (c) **Monte Carlo Simulation** of 10,000 portfolios within a sector to know which stocks are performing the best
- (d) Implementation of Mean-Variance Optimization Model to calculate the **Efficient Frontier** and the optimized portfolio for a target return (=20%)
- (e) Performance of the optimized portfolio on the 2023 stock market data and enhancement of returns with **dynamic rebalancing**
- (f) Comparison of the returns of the optimized portfolio with **NIFTY 50 Benchmark Index**



Project Methodology

1. Choosing the stocks

Among the various sectors in National Stock Exchange (NSE), Bombay, six sectors are chosen that exhibit diversity in the Indian Stock Market:

Auto, Banking, Metal, Fast Moving Consumer Goods (FMCG), Pharmaceuticals, and Information Technology (IT).

The NSE produces monthly reports identifying the stocks in each sector that have the most significant impact on the respective sector.

Based on NSE's report of Jan 2022, the eight most critical stocks from each of the six sectors are identified.

Symbol	Security Name	Industry	Close Price	Index Mcap (Rs. Crores)	Weightage (%)
AMARAABAT	Amara Raja Batteries Ltd.	AUTOMOBILE	619.20	6134	1.13
ASHOKLEY	Ashok Leyland Ltd.	AUTOMOBILE	132.55	19066	5.52
BAJAJ-AUTO	Bajaj Auto Ltd.	AUTOMOBILE	3564.05	46417	8.38
BALKRISIND	Balkrishna Industries Ltd.	AUTOMOBILE	2337.05	18975	5.51
BHARATFORGE	Bharat Forge Ltd.	INDUSTRIAL MANUFACTURING	715.00	18621	5.48
BOSCHLTD	Bosch Ltd.	AUTOMOBILE	16533.90	14142	7.61
ECHERMPOT	Eicher Motors Ltd.	AUTOMOBILE	2643.50	50869	6.82
EXIDEIND	Exide Industries Ltd.	AUTOMOBILE	172.05	7971	1.47
HEROMOTOCO	Hero MotoCorp Ltd.	AUTOMOBILE	2724.30	55381	6.54
MRF	MRF Ltd.	AUTOMOBILE	72148.10	14994	2.77
M&M	Mahindra & Mahindra Ltd.	AUTOMOBILE	885.30	84794	15.68
MARUTI	Maruti Suzuki India Ltd.	AUTOMOBILE	8591.30	114271	21.13
TVSMOTOR	TVS Motor Company Ltd.	AUTOMOBILE	666.30	13538	2.50
TATAMOTORS	Tata Motors Ltd.	AUTOMOBILE	517.75	92858	17.16
TATINDIA	Tata Investments of India Ltd.	AUTOMOBILE	1695.35	16678	3.08

Symbol	Security Name	Industry	Close Price	Index Mcap (Rs. Crores)	Weightage (%)
BRITANNIA	Britannia Industries Ltd.	CONSUMER GOODS	3535.30	41780	5.65
COLPAL	Colgate Palmolive (India) Ltd.	CONSUMER GOODS	1424.50	19010	2.57
DABUR	Dabur India Ltd.	CONSUMER GOODS	536.75	31354	4.24
EMAMILTD	Emami Ltd.	CONSUMER GOODS	498.30	10202	1.36
GODREJCP	Godrej Consumer Products Ltd.	CONSUMER GOODS	888.30	33853	4.55
HINDUNILVR	Hindustan Unilever Ltd.	CONSUMER GOODS	2373.75	202848	27.42
ITC	ITC Ltd.	CONSUMER GOODS	220.20	192501	26.03
MARICO	Marico Ltd.	CONSUMER GOODS	482.55	24978	3.38
NESTLEIND	Nestle India Ltd.	CONSUMER GOODS	18521.55	66021	8.93
PGHE	Procter & Gamble Hygiene & Health Care Ltd.	CONSUMER GOODS	14754.10	13907	1.88
RADICO	Radico Khurda Ltd.	CONSUMER GOODS	1075.85	8640	1.17
TATACONSUM	Tata Consumer Products Ltd.	CONSUMER GOODS	727.30	43625	5.90
UBL	United Breweries Ltd.	CONSUMER GOODS	1619.20	11574	1.56
MCDOWELL-N	United Spirits Ltd.	CONSUMER GOODS	866.40	25846	3.49
VBL	Varun Beverages Ltd.	CONSUMER GOODS	904.85	13732	1.86

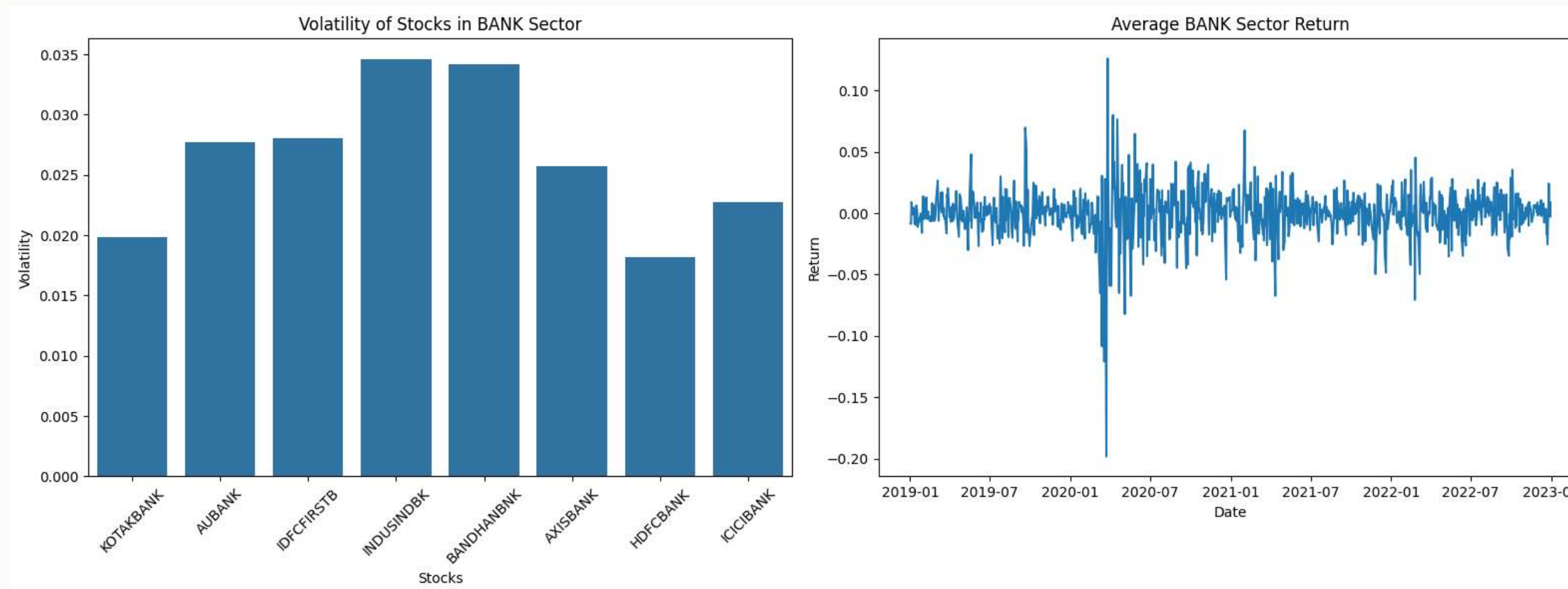
Most Critical Stocks in the sector of Auto and FMCG, as per NSE



Project Methodology

2. Sector Analysis

Various statistics of the stock data are analysed. **Daily log returns** of each stock are calculated, using the shift function. The log returns are also calculated for the sectoral indices for each sector, to analyse the sector as a whole. The **standard variation of the log returns are calculated to obtain the volatility** within the sector. Average returns are also calculated to review the performance of the sector.



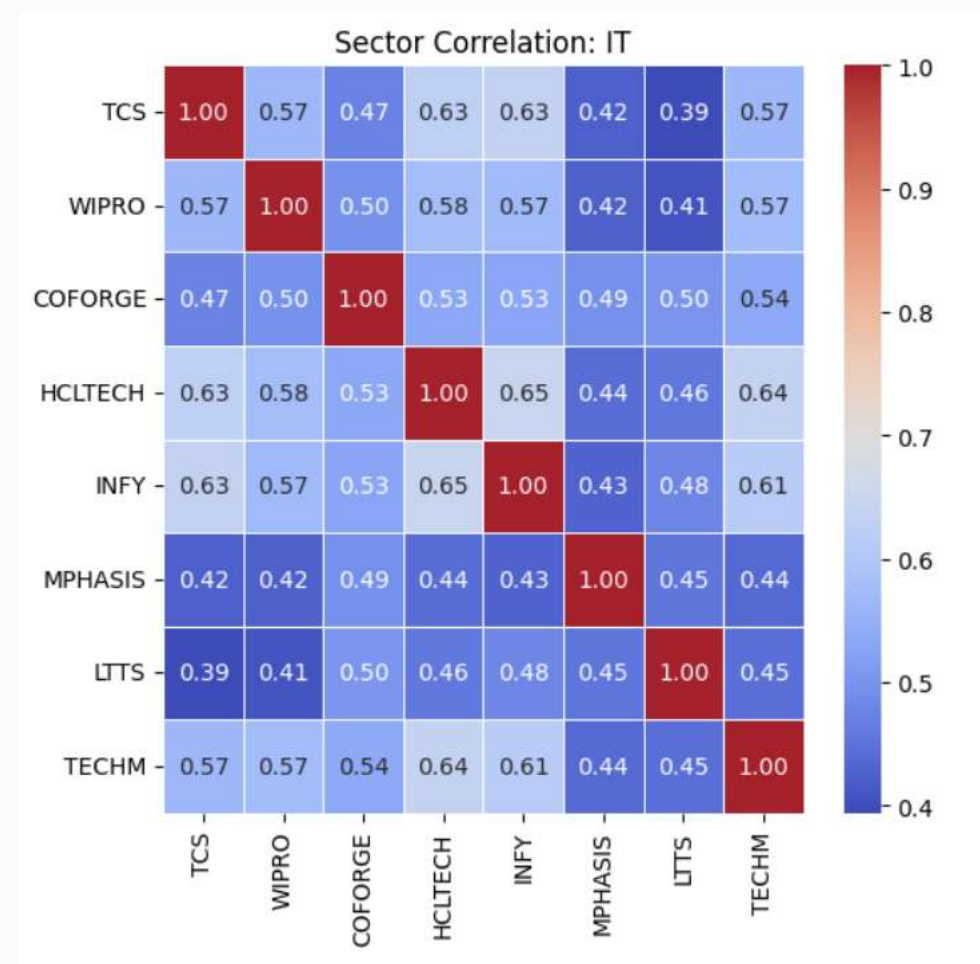
Volatility of each stock and Average Return of the BANK sector



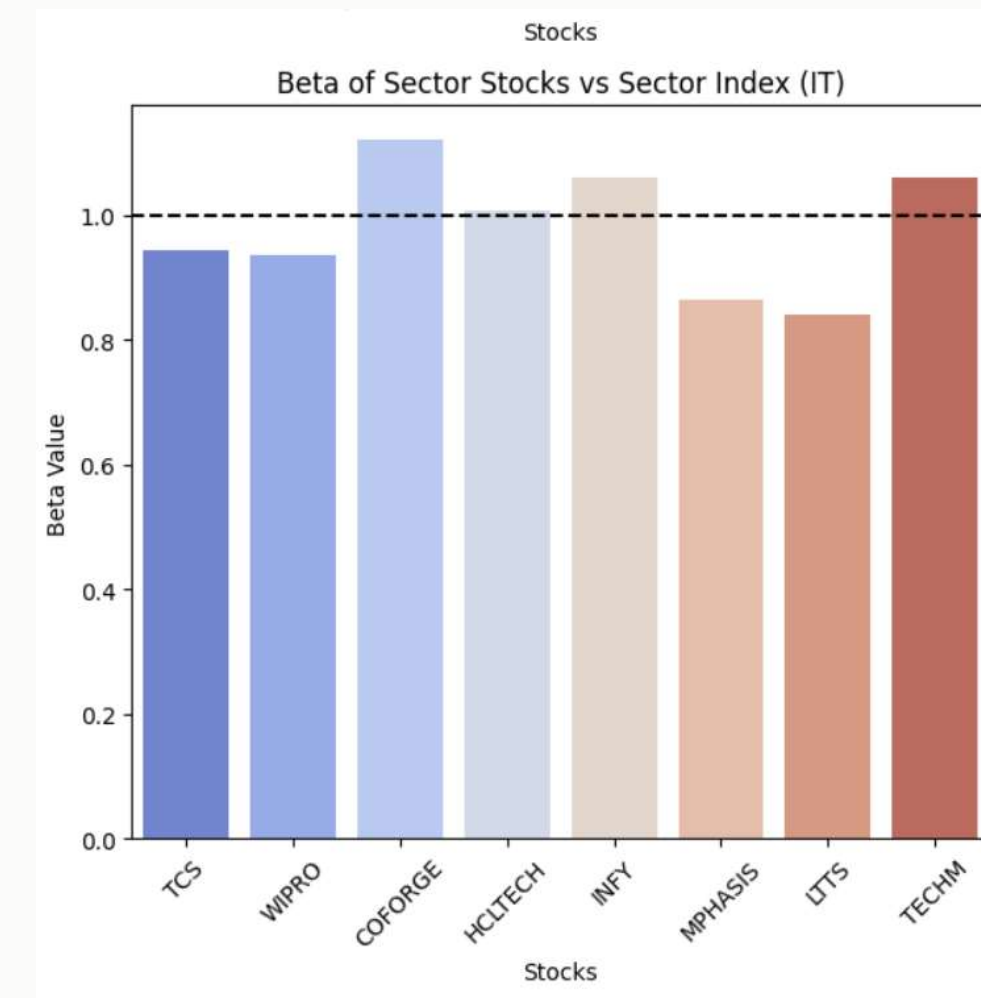
Project Methodology

2. Sector Analysis

Correlation matrix of the stock return data is calculated per sector, to analyse how strongly are the prices of a certain stock depend upon any other stock in the sector. This helps identify, which stocks to be put in the portfolio. The **beta values** for each stock is also calculated. It indicates how much the stock's price tends to move relative to the market.



Heat Map showing the correlation within the stocks of IT Sector



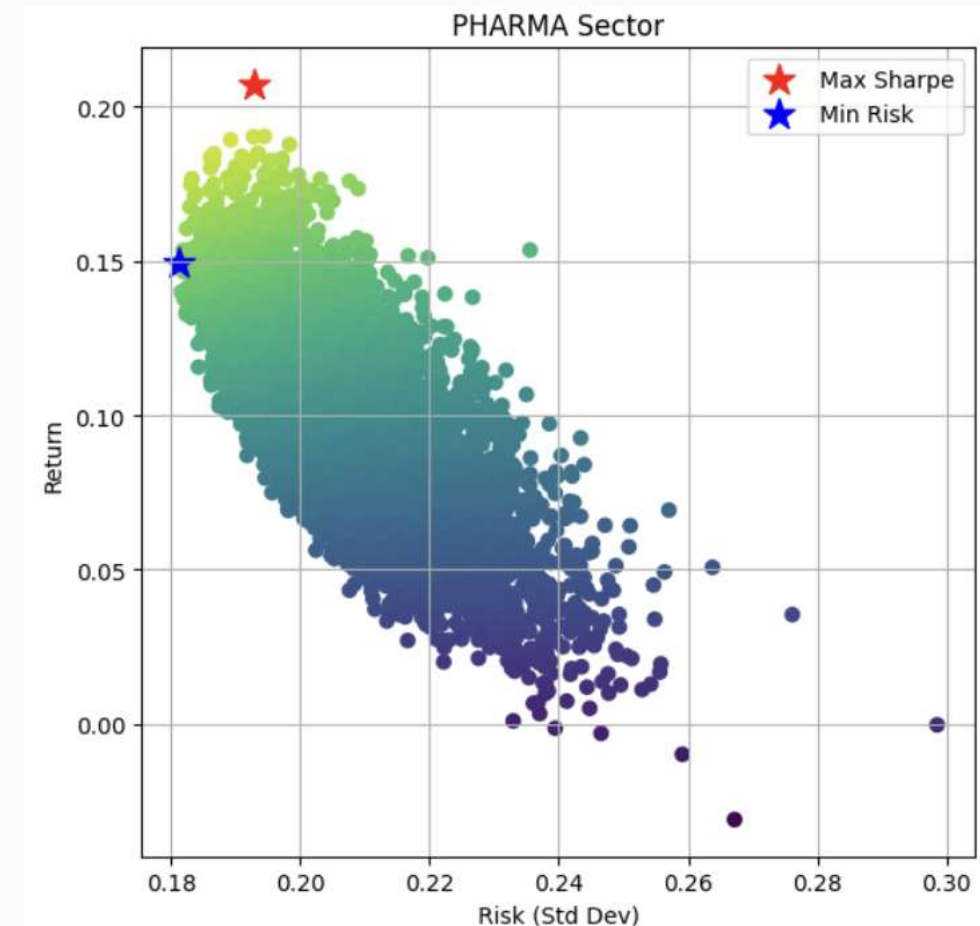
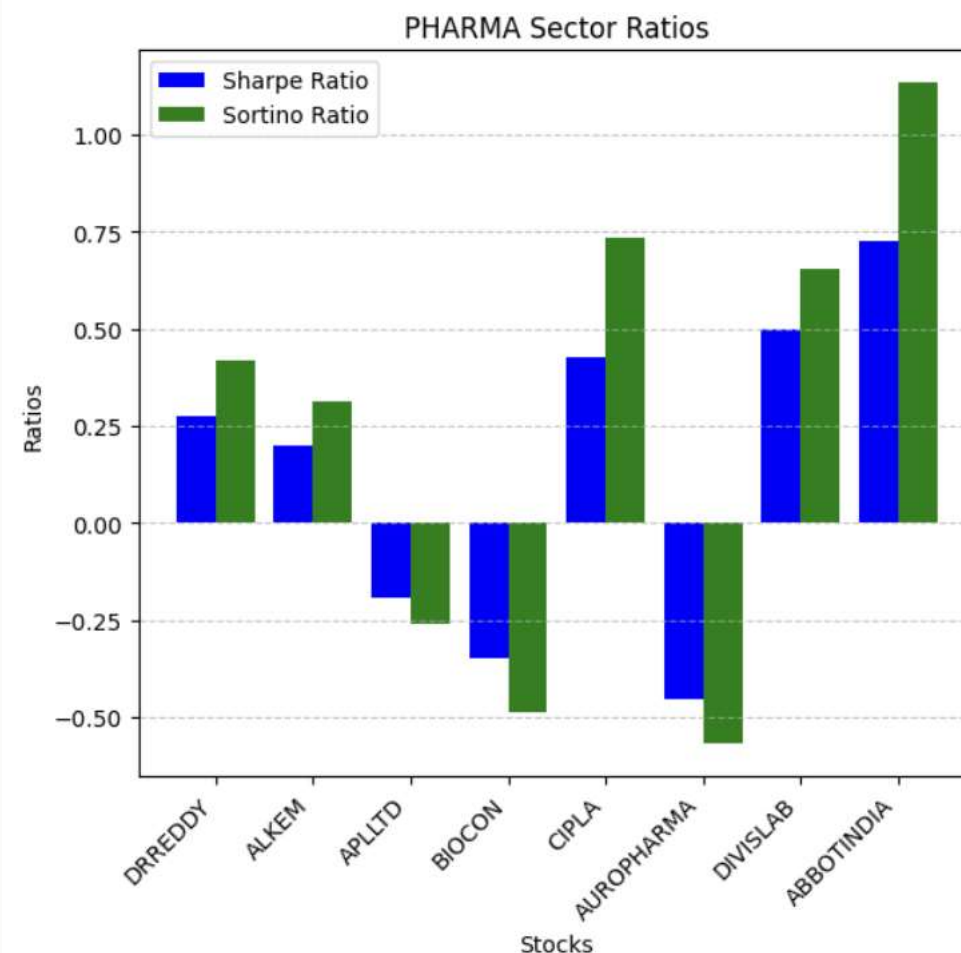
Beta Values of the stocks in the IT Sector



Project Methodology

3. Implementation of MVO Model

Sharpe and Sortino Ratios for each sector are calculated first. The risk free return rate is assumed to be 7% for the same (*Source: www.people.stern.nyu.edu*) Monte Carlo Simulations (10000) for all the stocks in a sector are also provided for review. The stocks shortlisted for the portfolio based on the insights provided by these ratios and Monte Carlo Simulations are: **'BALKRISIND', 'BAJAJ-AUTO', 'ICICIBANK', 'HDFCBANK', 'HINDUNILVR', 'INFY', 'HCLTECH', 'ADANIENT', 'APLAPOLLO', 'ABBOTINDIA'**



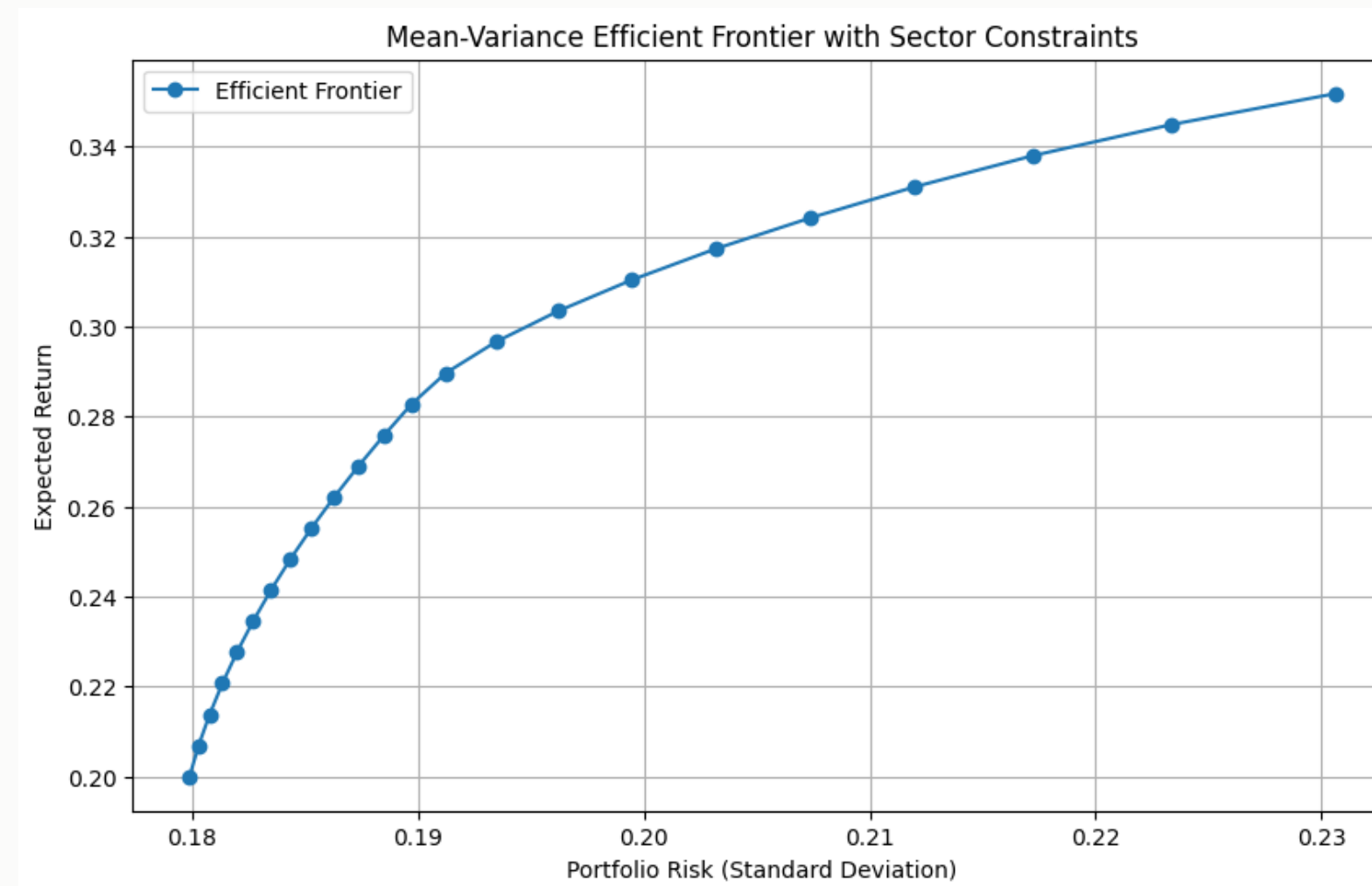
Plots showing the Sharpe and Sortino Ratios and Monte Carlo Simulations for PHARMA sector



Project Methodology

3. Implementation of MVO Model

Now, the MVO model is implemented on the shortlisted stocks. Sector based constraints and base constraints are specified. And, **the optimization problem, to minimize the risk given the constraints is solved** for every target return. The efficient frontier is built using the optimal portfolio's risk vs return analysis. The optimal portfolio for **20% return** is also shown as a pie-chart, which will be used for back-testing.



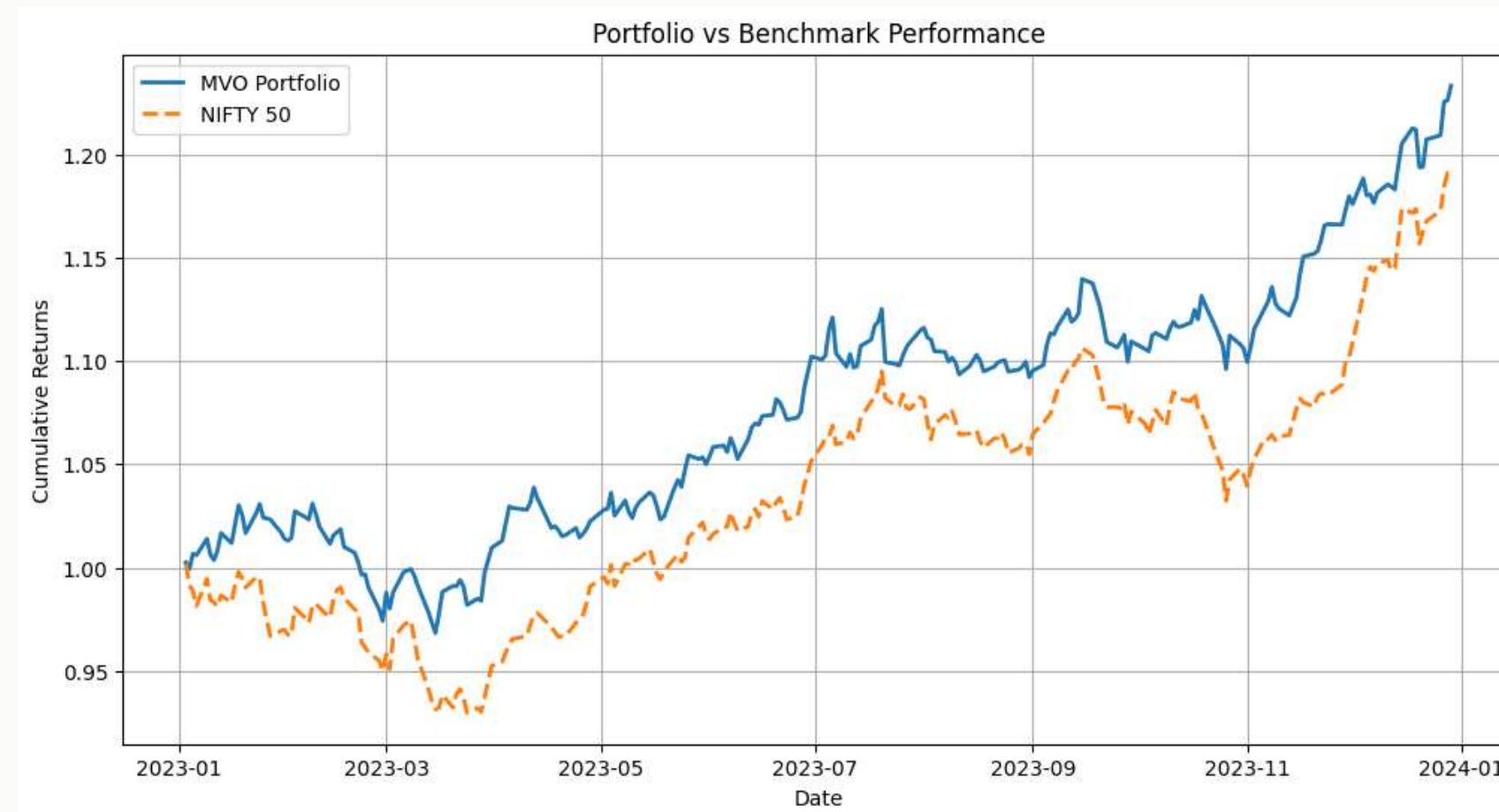
Efficient Frontier obtained by performing the MVO model analysis



Project Methodology

4. Back-testing

The optimal portfolio (for 20% return) is performed in the stock market during the 2023 year. The annual returns, volatility, Sharpe ratio and maximum drawdown is calculated for the year. And its performance is compared with the **NIFTY 50 Benchmark Performance**. It turns out, our portfolio did significantly better than the benchmark performance.



NIFTY Annual Return Rate: 19.14%

Our portfolio's Return Rate: 23.69%



Project Methodology

4. Back-testing

The portfolio is also dynamically rebalanced **half quarterly, quarterly, and monthly**, to obtain better returns. The process is run on a python file and results are shown using Dash and Plotly. When rebalanced monthly, the portfolio gains maximum returns, **(=26%)** against the 19% return of NIFTY 50 Index.

Thus, the portfolio obtained from the Mean-Variance Optimization Model, satisfactorily gives returns in the Year 2023.

Note: Please run the Jupyter notebooks in the order: *Data_Preprocess.ipynb*, *Sector_Analysis.ipynb*, *MVO_Model.ipynb*, *BackTesting.ipynb*, *monthwise_dashboard.py*. Also, please ensure that all the folders of stock data and csv files are in the same folder as those of the notebooks, while running them. Sorry for the mess :)



Conclusion

This project has presented an approach to portfolio optimization using the **Mean-Variance Optimization Model**, using stocks that performed the best in their respective sectors. Six sectors are chosen and analysed, and the stocks that were stable and gave maximum returns were shortlisted. The stock data from **January 1, 2019 to December 31, 2022** is used for the analysis. The optimal portfolio is prepared by incorporating the sector constraints and minimising risk for given target returns. The data from **January 1, 2023 to December 31, 2023** is used for judging the performance of the optimal portfolio. The study revealed that including stocks from different sectors made it diversified and hence, the risk was reduced to an extent. Also, the **Sharpe ratio and Sortino ratio** prove to be excellent metrics to nit pick the stocks to be included in the portfolio.

The further work includes incorporating sector related data, such as **USD-INR rates, Crude Oil Prices**, etc. into account, and dynamically choose the stocks from the sectors which are positively affected from these changes.

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References



References

- [1]** Jaydip Sen and Abhishek Dutta, *Design and Analysis of Optimized Portfolios for Selected Sectors of the Indian Stock Market, 2022.*
- [2]** Gupta, R. and Parikshit K. Basu, *Portfolio optimisation in the Indian stock market - industry sector analysis, 2022.*
- [3]** NSE Website: *<http://www.nseindia.com>*
- [4]** Historical data on stocks and sectors between the dates January 1, 2019 to December 31, 2024, from Yahoo Finance.
- [5]** Necessary Python modules like *yfinance*, *pandas datareader* and tools like Quandl, Dash and Plotly.
- [6]** Online AI models like ChatGPT